PROTECTING THE EGG DESIGN PROCESS

PROJECT OVERVIEW

The aim of the Egg Drop Challenge was to design a structure that protects a raw egg when dropped from height. The structure could only be built using straws and string. Materials such as glue, tape, foam, or paper were not allowed.
Because of these limitations, the structure had to rely on structural geometry, tension balance, and force distribution.
During the design process I developed two different prototypes. The first design was created independently, while the second design was developed with the help of AI suggestions.
The goal was to understand how structural design decisions influence the ability of the system to absorb impact forces.

TIMELINE

Understanding the challenge: I understood the rules and limits of the assignment.

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First design: I built a triangle-like structure and the egg did not break.

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AI-supported redesign: I developed a second design with AI support.

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Testing: I tested both designs by dropping them from height.

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Failure analysis: I observed that the second design failed and the egg broke.

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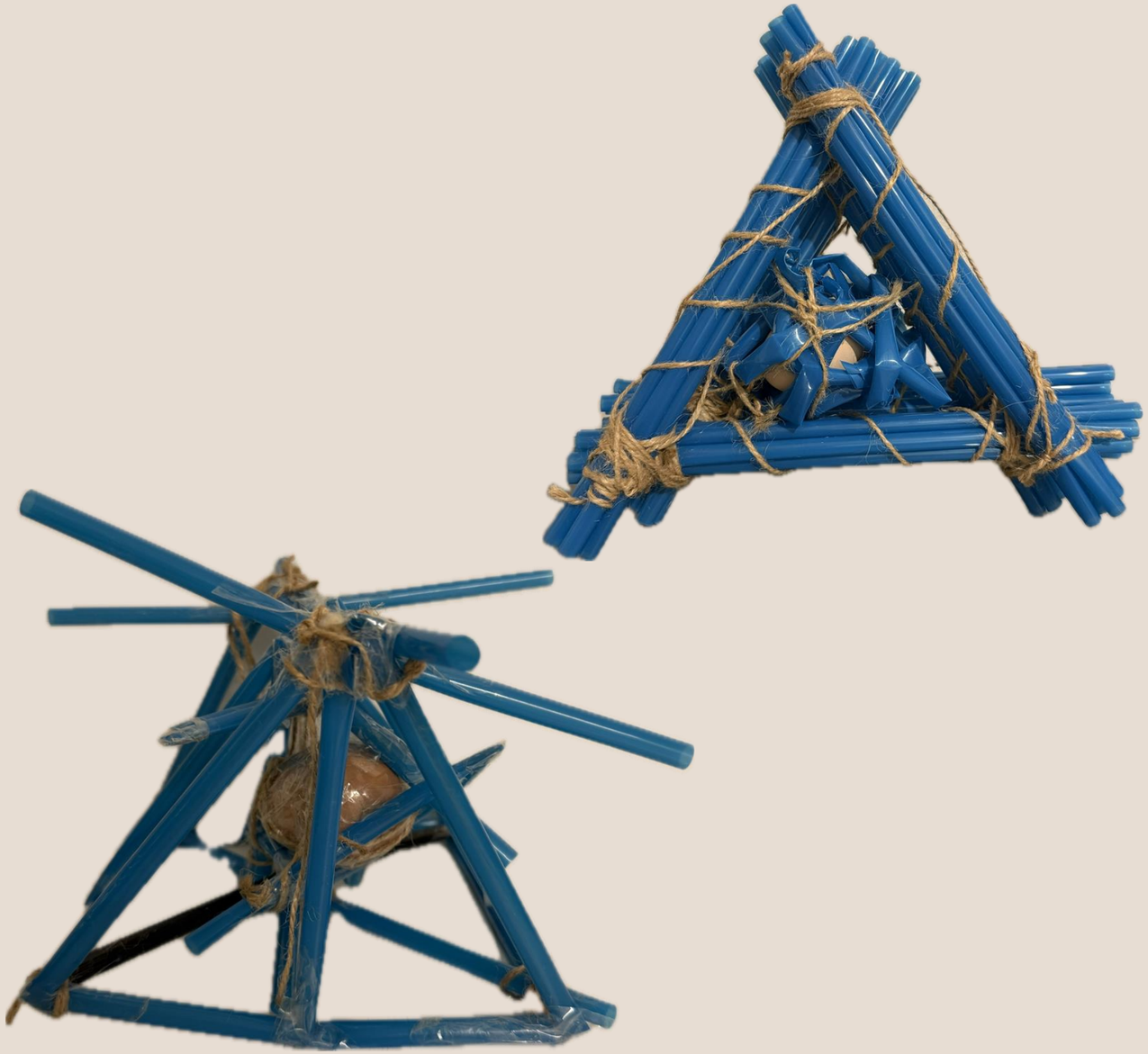
Reflection: I reflected on why the first design worked better.



TESTING AND FAILURE ANALYSIS

First Design

The first structure had a triangle-like geometry. This shape helped distribute the impact force through the structure.

Result:

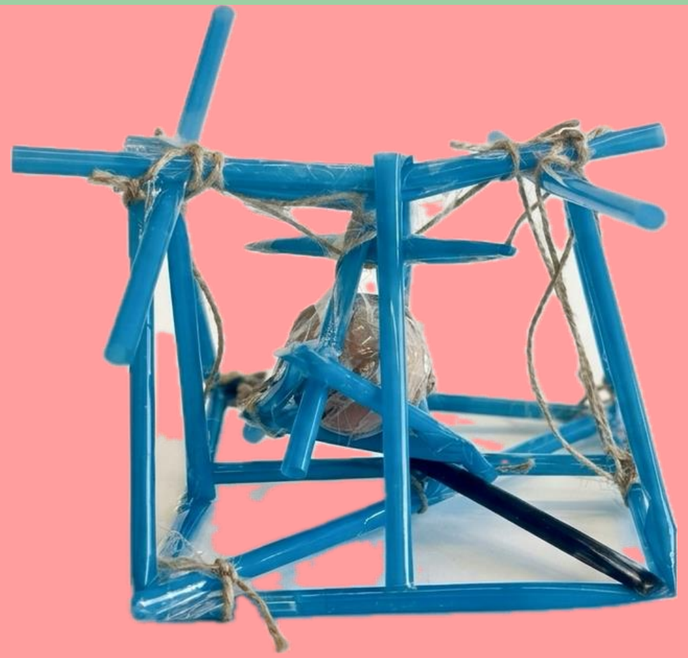
The egg did not break, which showed that the triangular structure was stable.

Second Design (AI-assisted)

The second design was developed with AI suggestions to explore alternative structural ideas.

Result:

During testing, the egg broke, which showed that the structure was less effective in absorbing impact.



FAILURE ANALYSIS

Center of Gravity

The egg needed to stay close to the center of the structure to maintain balance.

Impact Force Travel

When the structure hit the ground, the force traveled through the lower structural members.

First Element That Failed

Some structural members lost stability during the impact.

Failure Type

The failure happened at the member level, when individual straws bent or moved.

Prompts & AI Response

Prompt 1

"Design a strong egg drop structure that can protect an egg from a high drop."

AI Response Summary

AI suggested structural designs where the egg is positioned in the center so that the impact force is distributed through the frame.

- Clarifier – AI explained structural concepts such as force distribution.
- Expander – AI suggested alternative design possibilities.
- Re-framer – AI helped me rethink how impact should be absorbed.
- Critique – AI helped analyze why the second design failed.

Prompt 2

"How does impact force travel in a straw structure?"

AI Response Summary

AI explained that when the structure hits the ground, the force moves through the structural members. If the structure is symmetrical, the force spreads across the system.

Prompt 3

"Where should the egg be placed in an egg drop design?"

AI Response Summary

AI suggested placing the egg near the center to maintain stability and reduce direct impact.

Design Decisions & Transformations

Design Decisions

During the project I made several important decisions:

- placing the egg close to the center of the structure
- using geometry to distribute the impact force
- building a lightweight but stable structure

What I Accepted

I accepted AI explanations about:

- force distribution
- center of gravity
- structural balance

These ideas helped improve my understanding of the design.

What I Rejected

Some AI design suggestions were too complex to build using only straws and string.

What I Transformed

Some AI ideas were simplified and adapted to match the material limitations of the assignment.

FINAL REFLECTION

This project showed me that design is an iterative process. The first prototype worked better because its geometry was more stable.

Even though the AI-assisted design failed, the process helped me understand how structural geometry affects impact resistance.

Testing and failure analysis were important parts of learning from the design process.